Scheme and Answer Key for 25901920

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (Honours)/MGU-BBA (Honours)/MGU-BCA (Honours) Special Examinations, July 2026 (for 2024 Admission Regular Examinations, October 2025)

Third SEMESTER

Discipline Specific Elective (DSE) - MG3DSEPHY204 - INTRODUCTION TO MATERIALS SCIENCE (2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 Minutes **Maximum Marks: 50**

Reference: *Materials Science and Engineering – An Introduction*, William D. Callister & David G. Rethwisch (10th Edition)

Part A (2 Marks each)

1. Relation of axes and angles in the rhombohedral system (2 Marks)

- Three axes of equal length (1)
- Three equal angles ($\alpha = \beta = \gamma \neq 90°$) (1)

2. One application of Miller indices (2 Marks)

- Identification of crystal planes (1)
- Determination of slip planes/growth planes/XRD indexing (1)

3.Compare any two properties of ceramics and polymers (2 Marks)

Any two comparisons:

Ceramics	Polymers
High hardness	Low hardness
Brittle	Flexible
High melting temperature	Low melting temperature
Electrical insulator	Generally, insulator

(1 mark for each correct comparison)

4. Force vs interatomic separation graph (2 Marks)

Award marks as:

- Attractive force curve (0.5)
- Repulsive force curve (0.5)
- Net force curve (0.5)
- Equilibrium distance (r_0) marked where net force = 0 (0.5)

5. Solvus line (2 Marks)

- Boundary separating single solid solution from two-solid-phase region (1)
- Indicates maximum solid solubility (1)

6. Zone refining (2 Marks)

- Purification technique using moving molten zone (1)
- Impurities concentrate in molten region and move to one end (1)

7. Relation between transformation temperature and microstructure (2 Marks)

- Lower transformation temperature → finer microstructure (1)
- Higher transformation temperature → coarser microstructure (1)

8. Phase transformation and significance (2 Marks)

- Change from one phase to another (1)
- Alters properties such as strength, hardness or ductility (1)

9. Heterogeneous nucleation (2 Marks)

- Nucleation occurring at surfaces/interfaces/impurities (1)
- One suitable example (1)

10. Why heterogeneous nucleation occurs more readily? (2 Marks)

- Lower activation energy (1)
- Existing surfaces reduce surface energy barrier (1)

11. Two techniques to improve oxidation resistance (2 Marks)

- Alloying (Cr, Al, Si)
- Protective coatings
- Surface treatments
- Oxide film formation (1 mark each)

12. Importance of ductility (2 Marks)

- Allows plastic deformation before fracture
- Wire drawing/forming
- Improves toughness
- Provides warning before failure (1 mark each)

13. Why ductile fracture is preferred? (2 Marks)

- Gives noticeable plastic deformation before failure (1)
- Prevents sudden catastrophic fracture (1)

14. Thermal fatigue (2 Marks)

- Failure due to repeated heating and cooling cycles (1)
- Caused by cyclic thermal stresses (1)

Part B (5 Marks each)

15. Compare ionic and covalent bonding (5 Marks)

- Ionic bonding explanation – 1
- Covalent bonding explanation – 1
- Bond energy comparison – 1
- Melting point comparison – 1
- Electrical conductivity comparison – 1

16. Draw and explain (111) plane and [111] direction (5 Marks)

- Correct cubic unit cell (1)
- Correct (111) plane (2)
- Correct [111] direction (1)
- Explanation/definition (1)

17. Effect of impurities on diffusion (5 Marks)

- Definition of diffusion (1)
- Influence of impurities (2)
- Suitable example(s) (1)
- Conclusion (1)

18. Phase diagram comments (5 Marks)

(a) Importance of phase diagrams (2.5)

- Melting
- Solidification
- Casting
- Heat treatment
- Alloy design

(b) Solubility limit (2.5)

- Solubility limit explained (1)
- Formation of second phase beyond limit (1)
- Suitable example (0.5)

19. Gibbs Phase Rule (5 Marks)

- Equation $F = C - P + 2$ (1)
- Binary system ($C = 2$) (1)
- Single-phase region (1)
- Two-phase region (1)
- Invariant point explanation (1)

20. Polymer degradation mechanisms (5 Marks)

- Thermal degradation
- Oxidative degradation
- Photodegradation
- Hydrolytic degradation
- Biological degradation (1 mark each)

21. Stress–strain graph showing nonlinear elastic deformation (5 Marks)

- Correct graph (2)
- Linear region identified (1)
- Nonlinear elastic region explained (1)
- Recovery after unloading explained (1)

22. Caterpillar analogy for dislocation motion (5 Marks)

- Define dislocation (1)
- Describe caterpillar movement (1)
- Explain movement of dislocation (2)
- Advantage: lower energy required than simultaneous slip of all atoms (1)